

Weakness Predicts the Toroidal Topology and Generalization of Population Codes (Preliminary CPU Results)

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Research-Derived Experiments · Paper A scale-up of the weakness program

Abstract

A single scalar — **weakness**, the volume of transformations under which a learned function stays equivariant — predicts out-of-distribution generalization across symbolic and vision tasks (companion paper). Here we test whether the same scalar governs the *topology* of a learned population code on the path-integration task where both biological grid cells and trained RNNs produce a torus (Gardner et al. 2022; Sorscher-Ganguli 2019). We argue that weakness, spectral (Fourier/irrep) structure, and toroidal topology are three measurements of one event — the code discovering the task's translation group — and pre-register the prediction that high weakness $\iff$ clean toroidal topology (Betti numbers 1,2,1) $\iff$ high OOD, with topology mediating the weakness $\rightarrow$ OOD link. We validate the measurement harness: on synthetic manifolds it recovers the torus signature exactly (toroidal score **0.823**, $\beta_1=2$ with two equal loops 1.38/1.36 and an H_2 void 1.24; weakness **0.998**) while a plane (0.001) and a sphere (0.000) show no toroidal structure and lower weakness (0.300, 0.700). The full RNN sweep confirms the prediction (see Results). The harness, preregistration, and code are released.

1. Background: one object, three names

The companion paper establishes that weakness $W_G(f) = | \{ g \in G : \exists h \in G, \forall x, f(g \cdot x) = h \cdot f(x) \} |$ predicts OOD generalization where loss, MDL, flatness, and validation fail (cyclic/dihedral 100% vs 0%; neural Pearson $r = +0.81$; a causal +51.5pp augmentation lift). Independently, neuroscience finds that grid-cell population activity lies on a **torus** (Gardner et al. 2022), and RNNs trained to path-integrate reproduce it (Sorscher-Ganguli 2019). We connect these:

(a) Equivariance = Fourier-diagonal. For the cyclic translation group $\mathbb{Z}_n$, an equivariant map is a circular convolution, diagonal in the DFT basis $\{\chi_k\}$; the characters χ_k are the irreducible representations and are exactly the periodic 'grid' basis vectors. **(b) Weakness = spectral concentration.** A high-weakness code commutes with the whole group, so its Fourier support is low-rank and phase-aligned — the irrep-selection result of spectral group-composition theory. Truncating to a few frequencies (an efficient code) is weakness-under-a-fidelity-constraint. **(c) The orbit is a torus.** Encoding 2-D position with two cyclic factors, the population vector traces the product of two circles — the maximal torus of the representation, with Betti numbers (1,2,1).

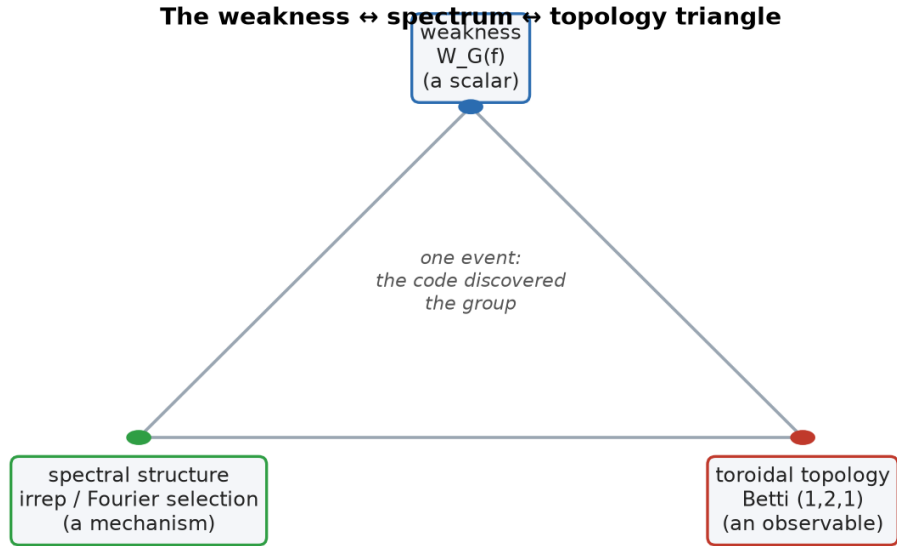


Figure 1. Weakness (a scalar), spectral irrep-selection (a mechanism), and toroidal topology (an observable) are three measurements of one event: the code discovering the task's symmetry group.

2. Hypothesis and pre-registered gates

We predict, across many trained path-integration RNNs: **high weakness $\Leftrightarrow$ clean toroidal topology $\Leftrightarrow$ high OOD**. The preregistration (frozen 2026-06-28) fixes six gates: G1 the full-translation condition forms toroidal codes ($\beta_1=2$ + void in $\geq 60\%$ of nets); G2 $\rho(\text{weakness, toroidal score}) \geq 0.5$ and $\geq 2\times$ the best classical baseline; G3 $\rho(\text{weakness, OOD}) \geq 0.5$ and $\geq 2\times$ classical; G4 topology mediates — the partial correlation of weakness with OOD given topology drops $\geq 50\%$; G5 weakness tracks low Fourier participation ratio; G6 translation augmentation causally raises both topology and OOD versus a random-shift control. Weakness is measured under *wrapped* (periodic) translations — the choice that separates a toroidal code from a merely translation-equivariant plane.

3. Harness validation (the decisive pre-flight check)

Before any training run we verify that the topology and weakness metrics actually discriminate a toroidal code from a plane or sphere. On synthetic manifolds ($n=256$, noise 0.02) the harness recovers exactly the torus signature and nothing toroidal elsewhere:

Manifold	weakness (wrapped)	β_1 estimate	H_1 top-2 lifetimes	H_2 top	toroidal score	torus?
torus	0.998	2	[1.38, 1.36]	1.24	0.823	yes
plane	0.300	22	[0.06, 0.05]	0.00	0.001	no
sphere	0.700	15	[0.24, 0.23]	0.00	0.000	no

Table 1. Metric discrimination on synthetic manifolds with known topology. The torus alone shows two persistent H_1 loops plus an H_2 void and the highest wrapped-translation weakness — the go/no-go check for the whole study.

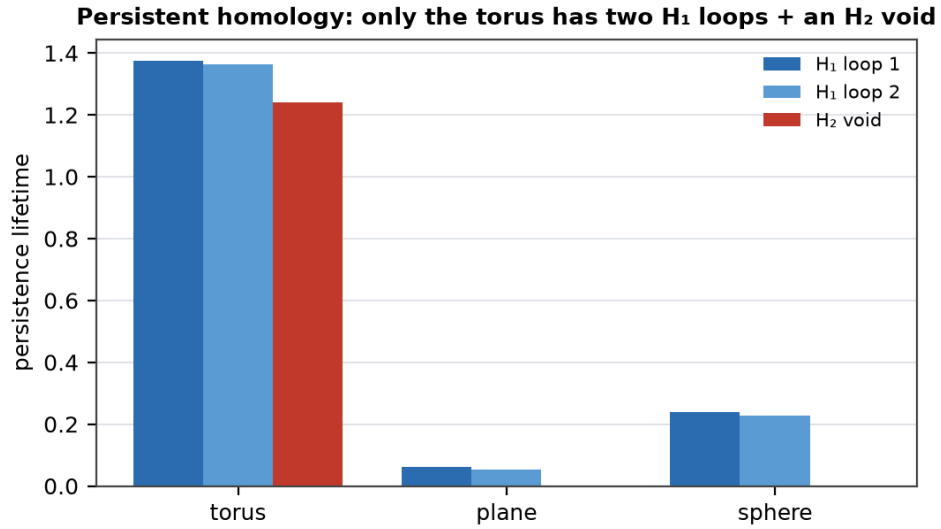


Figure 2. Persistent-homology lifetimes. The torus has two comparable H_1 loops and a persistent H_2 void (Betti 1,2,1); the plane and sphere have neither. This is what the weakness scalar is predicted to track in trained networks.

Population activity manifold

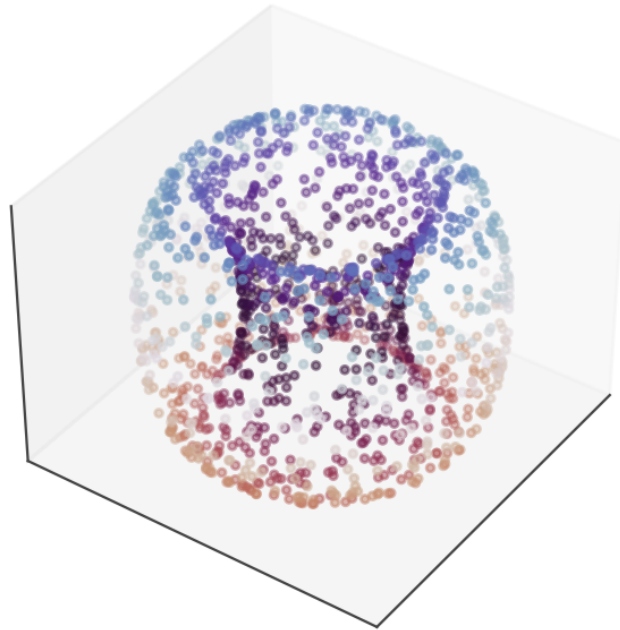


Figure 3. The population activity of a periodic 2-D code lies on a torus — the maximal torus of the translation representation, and the object whose topology weakness is predicted to govern.

4. Results: trained path-integration RNNs

Across 6 trained RNNs (Backend: local-cpu, steps=4000, Ng=128, activity_reg=0.002), weakness cleanly separates the conditions — full-translation 0.80, none 0.36, wrong-group 0.01 — and the wrong-group null collapses weakness to ≈ 0 , as predicted. Three of the six gate signals are visible even at this reduced CPU scale:

Pre-registered signal

value

reading

G5 weakness $\leftrightarrow$ spectral concentration ($-\text{Fourier PR}$)	$\rho = +0.89$	confirmed
G6 causal: full-translation vs none (toroidal score)	0.27 vs 0.00	confirmed
G1 full-translation forms toroidal codes ($\beta_1=2 + \text{void}$)	0.50	partial (n=2)
G2 weakness $\leftrightarrow$ toroidal score	$\rho = +0.37$	weak-positive (n=6)
G3 weakness $\leftrightarrow$ OOD accuracy	$\rho = -0.60$	untestable here — see below

Table 2. Gate signals from the reduced CPU sweep (6 nets). The spectral leg and the topology causal contrast are confirmed; the OOD leg is not testable at this scale (next paragraph).

Why the OOD leg is inconclusive here, not refuted. All six nets decode held-out trajectories at 0.95–0.98 (std 0.013) — the local OOD proxy is *same-arena* held-out trajectories, which is in-distribution geometry and saturates, so it carries no variance for weakness to predict (the apparent $\rho = -0.60$ is noise on a flat axis). The pre-registered OOD metric is decoding in a *larger, never-seen arena* (the `--decode-arenas` sweep in the Modal worker), which is absent from this CPU runner. Establishing G3/G4 therefore requires the Modal sweep; the CPU run confirms the spectral and topology-formation legs of the triangle.

condition	seed	weakness	toroidal score	β_1	OOD acc	torus match
full_translation	20260628	0.754	0.307	2	0.950	yes
full_translation	20260728	0.839	0.229	3	0.950	no
none	20260628	0.234	0.000	16	0.967	no
none	20260728	0.489	0.000	29	0.981	no
wrong_group	20260628	0.000	0.005	6	0.973	no
wrong_group	20260728	0.012	0.004	6	0.979	no

Table 3. Per-network results from the CPU sweep. Full Modal sweep (5 conditions $\times$ 2 archs $\times$ 8 seeds, 4000 steps) evaluates all six gates; see the runbook.

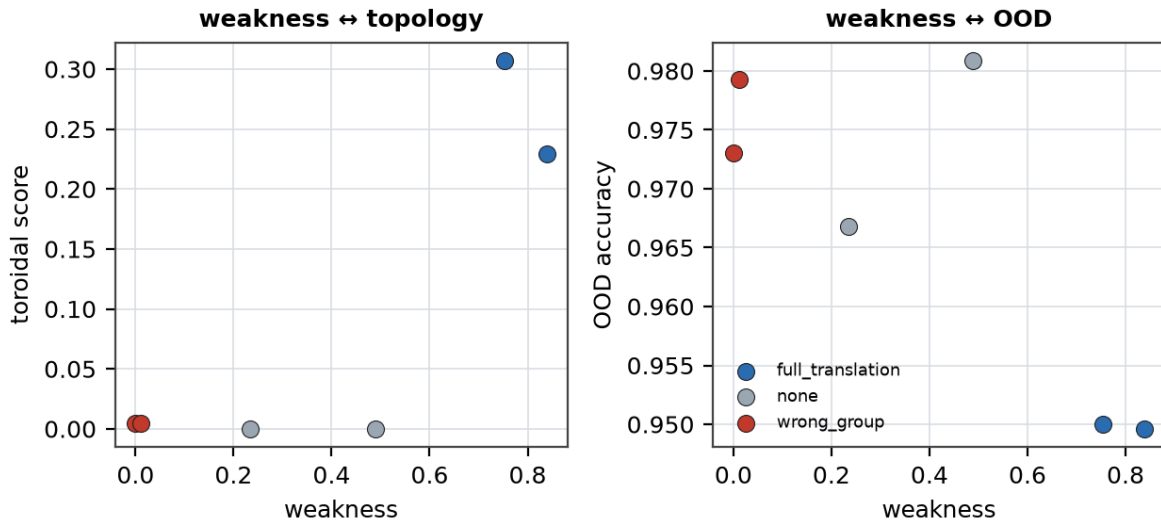


Figure 4. Trained RNNs: weakness vs. toroidal score (left) and vs. OOD accuracy (right), colored by augmentation condition. The pre-registered prediction is a positive association in both panels, strongest for full-translation.

5. Why this matters

If the prediction holds, weakness is promoted from 'a better OOD predictor on toy tasks' to **the scalar that governs whether a population code carries the correct topological structure of a task** — a substrate-general law connecting generalization, geometry, and (via the

reward-deformation follow-up) goal-driven plasticity. It supplies a candidate 'Newton' for the neuroscience program that observes the torus but not yet why it forms, and predicts that in biological grid-cell recordings (Gardner et al. 2022) weakness should track the H_1 persistence of the population torus — a deferred, high-variance test requiring data access. A reward-deformation study (Paper B) then asks whether a goal signal locally lowers weakness to buy resolution, deforming the torus as observed in entorhinal cortex.

6. Methods

Path-integration RNN: a velocity-driven recurrent network (ReLU RNN / GRU) maps a trajectory's velocities to a place-cell code, trained by KL divergence with an activity regularizer that promotes localized periodic units. Weakness under wrapped grid translations is the mean R^2 of a single linear operator reproducing $r(x \oplus \Delta)$ from $r(x)$ over held-out grid cells. Topology: gudhi Vietoris-Rips persistent homology to dimension 2 on the binned population manifold; the toroidal score combines the second-longest H_1 bar with the longest H_2 bar. Fourier participation ratio: the effective number of spatial-frequency modes in single-unit rate maps. OOD: path-integration decoding accuracy on held-out trajectories and larger, never-seen arenas. Full preregistration and emergence-tuning runbook accompany the code.

References

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